

Getting up to speed: the OML descent residue

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Purpose. A zero-to-ready reading path for the one open question in the OML extension thread that needs *human* mathematical work. It assumes you are coming back cold. It does not re-derive the research note — it points you into it (`oml_extension_problem.tex/.pdf` and `verification/`) and gives you the “you are here” map the note assumes you already hold.

When you can answer the **checkpoint** questions at the end of each section without looking, you are ready for the question in §5.

Companion files.

- `oml_extension_problem.tex / .pdf` — the full research note (the authority; this doc is a guide to it).
- `verification/` — the small computational and hand-verification scripts.
- Memory: `project_overview.md`, `oml_relational_prob_novelty.md`, `oml_two_point_spaces.md`.

0. The one-sentence version

Build a σ -additive probability theory on a non-distributive orthomodular lattice **without descending from a sample space** — or prove it can’t be done — by supplying the missing σ -OML representation that all three known routes fail to give.

Everything below is the scaffolding needed to understand why that sentence is (a) precise, (b) genuinely open, and (c) worth the effort.

1 Prerequisites to know cold

You need four things at your fingertips. If any is shaky, fix it before §2.

1a. OML / OMP, and meet-zero vs. orthogonal — the gap that drives everything

- **Orthomodular lattice (OML):** a bounded lattice with an orthocomplement $a \mapsto a^\perp$ (involution, order-reversing, $a \wedge a^\perp = 0$, $a \vee a^\perp = 1$) satisfying the orthomodular law $a \leq b \Rightarrow b = a \vee (a^\perp \wedge b)$. Boolean = the distributive special case. Motivating non-Boolean example: $L(H)$, closed subspaces of a Hilbert space. Smallest: MO_3 (three complementary pairs pasted at 0 and 1). (*Def. in note §2, Defs 2.2–2.3; Hasse diagram Fig. 1.*)
- **The gap.** Two elements are *orthogonal* ($a \perp b$) if $a \leq b^\perp$; they are *meet-zero* if $a \wedge b = 0$. In a Boolean algebra these coincide. In a non-distributive OML, **orthogonal $\Rightarrow$ meet-zero but NOT conversely**. This single gap is the whole extension obstruction — internalise

it. MO_2 is the smallest concrete witness ($\{1, 2\} \wedge \{1, 3\} = 0$ but $\{1, 2\} \cap \{1, 3\} = \{1\} \neq \emptyset$).
(Note §2, “meet-zero versus orthogonal”; [verification/concrete_meetzero_vs_orthogonal.py](#).)

- **OMP / concrete logic.** An orthomodular *poset* only requires joins for orthogonal pairs. “Concrete” = “set-representable” = embeds in $(\mathcal{P}(X), \subseteq, \text{complement})$ preserving orthogonal joins. **Trap:** concrete $\neq$ Boolean (MO_3 is concrete); the dividing line that closes the meet-zero/orthogonal gap is *intersection-richness*, not concreteness. **Naming note:** the abstract condition $a \wedge b = 0 \Rightarrow a \perp b$ is Tkadlec’s “*Boolean orthoposet*” (Math. Bohemica 119, 1994) — which does **not** mean Boolean algebra. Use “intersection-richness” as the working term.

1b. The state / valuation / charge ladder

Three strengthenings of additivity, strictly increasing. Keep them distinct — collapsing them is one of the recurring errors on this thread.

1. **State:** $s(1) = 1$, additive on *orthogonal* pairs.
2. **Valuation** (local convention in the note): additive on *meet-zero* pairs (binary, strictly stronger).
3. **Extends to a (Boolean) charge:** the n -ary condition $\sum_i s(a_i) \leq 1$ for every pairwise-meet-zero family (strictly stronger again). *This is the extension axis.*

Sanity anchor: on MO_3 , $s \equiv \frac{1}{2}$ is a valuation but fails (3) at $\{p, q, r\}$ ($\frac{3}{2} > 1$). (Note §2; [verification/mo3_extension.py](#).)

1c. McDonald–Bimbó duality — the dual space $S_0(A)$ and its points $P(A)$

The relational realist move: instead of starting from a space of outcomes, build the dual from the lattice itself.

- $S_0(A)$ = the space of **filters** on A (compact, Stone-like).
- $P(A) \subseteq S_0(A)$ = the **principal filters** — the “physical points,” the realisation datum. *Canonical*, unlike the Boolean case where the pure points are a non-canonical choice.
- $h(a) = \{\text{filters containing } a\}$ is an OML isomorphism onto the $\perp$ -stable clopens. Set $\mu(h(a)) = s(a)$.
- **The duality is FINITARY.** $h(\bigvee_n a_n) = (\bigcup_n h(a_n))^{\perp\perp}$ holds for *finite* joins, fails for countable ones. This finitariness is the wall the whole open question runs into. (Note §2; [primary-source check in verification/mb_primeness_check.md](#).)
- Why MB and not the rival Cannon–Döring OML duality? MB carries $P(A)$ as explicit structure; both are purely structural (no measure), but MB’s filter points are exactly the realisation datum the programme needs.

1d. Loomis–Sikorski — the Boolean engine the OML case is missing

This is the keystone. In the Boolean world, the descent argument (“ σ -additive $\Leftrightarrow$ the measure concentrates on the physical points”) runs on a **Loomis–Sikorski** representation: a σ -complete Boolean algebra is a σ -tribe of sets mod a σ -ideal. **There is no OML analogue** — and supplying one (or proving none exists) *is the open question*. Hold this as: “the Boolean proof has an engine; the OML proof needs one and doesn’t have it.”

Checkpoint 1

- (i) Give the MO_2 witness for meet-zero $\neq$ orthogonal. (ii) Why does $s \equiv \frac{1}{2}$ on MO_3 clear the valuation bar but fail extension? (iii) What is $P(A)$, and why is it canonical here but not in the Boolean case? (iv) What does Loomis–Sikorski do in the Boolean descent proof,

and what’s the OML status of it?

2 The narrowing: from “the extension problem” to “the one residue”

The headline question — *when does a state on an OML A extend to a σ -additive measure on $S_0(A)$?* (note, Question 2.6) — splits into **two independent axes** that the Boolean case fuses. Most of the problem is already settled; one residue is live.

A state s on A faces two independent axes:

	EXTENSION axis	DESCENT axis
Question	Does μ extend to a finitely additive charge on the <i>full</i> Boolean clopen algebra of $S_0(A)$?	Once extended, does the measure concentrate on the physical points $P(A)$?
Governed by	the meet-zero/orthogonal gap (§1a)	σ -additivity — needs the missing engine (§1d)
Status	SETTLED: Horn–Tarski / Pitowsky feasibility; finite case = decidable LP; can fail for every state.	OPEN. <i>This is the residue.</i>

Two sharp results pin down the extension axis so you don’t re-fight it:

- **Finite case:** a decidable linear program. MO_3 shows it can fail for *every* state. Not a frontier. (Note §2, §5 “Extension axis”.)
- **Infinite $L(H)$, normal states — KILLED.** The clustering argument inside a single 2-plane: k distinct lines force $k \cdot s(e) \leq 1$ for all k , so any state with $s(e) > 0$ on some finite-dim e fails to extend. That’s *every normal (Gleason) state*. σ -completeness irrelevant; the argument is finitary. (Note §2.1, Prop 2.7; `verification/lh_infinite_extension.py`.)

So on the extension axis the *only* survivors are the **singular states** of $L(H)$ ($s(e) = 0$ on every finite-dim e) — see §5. The descent axis is where the real open problem lives — see §3.

Checkpoint 2

Reproduce the two-axis table from memory. Why is the extension axis “settled but partly degenerate”? Which states on $L(H)$ survive the clustering kill, and why does the kill not touch them?

3 The open question, stated crisply

Does a σ -complete, concrete, non-Boolean, infinite OML admit a Loomis–Sikorski-type representation (a σ -tribe of sets mod a σ -ideal), or a countable-join-preserving σ -Stone duality — or can one prove none exists?

A positive answer is the engine for *relational probability without realizations*: σ -additive probability built from the entailment relation of a non-distributive OML, **not** descended from any sample space — the non-Boolean analogue of localic (pointless) measure theory. A negative answer (an impossibility theorem) closes the whole cluster, including both residues, at once.

The outcome is **binary and large**. (Note §3 “finitary-to- σ bridge”; §5 “Descent axis”; “What is at stake”.)

Why this is the live stake, not just a loose end. The motivating idea (“probability from relations, point-free”) might *look* already delivered by $L(H) + \text{Gleason}$. It is not — and seeing why is the conceptual crux:

The (A)/(B) point-space equivocation — do not let this creep back

“Point-free” slides between two point-spaces. **(A)** the Hilbert rays $L(H)$ and the Gleason density ρ are *built from*; **(B)** the MB dual filters $P(A)$. The programme defines *realization* = concentration on **(B)**. Gleason removes **(B)** but **requires (A)** — and is a *representation* theorem (the most point-ful result available, reducing every lattice state to the point datum ρ). So $L(H)/\text{Gleason}$ witnesses the $\text{PR}_{\text{lat}} / \text{PR}_{\text{dual}}$ **separation**, NOT point-freeness, and is not the existence proof. Because it isn’t, the σ -OML representation (§3) is the *only* candidate engine. **Never write “Gleason already builds relational probability.”** (Memory: *oml_two_point_spaces.md*; note “What is at stake”.)

Checkpoint 3

State the open question in one sentence without looking. Explain why Gleason on $L(H)$ does *not* already answer it (the (A)/(B) distinction). What would a positive vs. negative answer each deliver?

4 What is already ruled out (don’t redo this)

Coming back cold, the danger is re-walking dead ground. The following are closed; cite them, don’t re-derive them.

- **Three named routes to a σ -OML engine — all blocked.** (Note §3.)
 - (i) **RDP / effect-algebra:** Loomis–Sikorski holds for σ -MV and monotone σ -effect-algebras *with* RDP, but lattice-EA has RDP iff it is MV, and $\text{OML} \cap \text{MV} = \text{Boolean}$. So **OML + RDP $\Leftrightarrow$ Boolean** — the machinery has no non-Boolean OML to act on.
 - (ii) **MacNeille completion:** the MacNeille completion of an OML need not be orthomodular (Harding 1991) — can’t complete to absorb joins.
 - (iii) **σ -Stone duality:** none exists. MB is finitary; Freytes gives only an equational theory, no set/tribe representation.
- **Normal states on $L(H)$ don’t extend.** Clustering argument (§2); hand-verified dichotomy in *verification/lh_singular_dichotomy.md*.
- **Both standing negatives stress-tested against primary sources (MISS, 2026-06-04).** (i) No impossibility theorem for the *concrete* class — the load-bearing word is “concrete”: the one positive occupant $P(H) + \text{Gleason}$ is *non-concrete* (Kochen–Specker $\Rightarrow$ no separating two-valued states). (ii) No σ -Loomis–Sikorski: both OML dualities take infinite joins as a *closure* not a union (Cannon–Döring $\text{cls}(\bigcup S_i)$, MB $(\bigcup h(a_n))^{\perp\perp}$). *Overclaim guard:* Gudder concrete logics ARE set-representable non-Boolean — but point-ful posets, not a σ -LS theorem.
- **Prior art map (scout, PARTIALLY OCCUPIED).** Every existing construction gives up exactly one needed property: Döring 2009 (finitely additive), Bohrfication/HLS (σ -additive but via a *distributive* internal locale, per-Boolean-context), localic valuations

(distributive by definition), Gleason (not point-free). The unoccupied conjunction — point-free $\times$ σ -additive $\times$ natively non-distributive $\times$ directed-system-built — is exactly the residue. (*Note §4; memory oml_relational_prob_novelty.md.*) The directed-context- $\Rightarrow$ -non-distributivity *idea* is NOT new (it’s the shape of Bohrfication and Gunji et al.’s colimit generation); novelty lives entirely in the σ -additive-point-free-native combination.

Checkpoint 4

Name the three blocked routes and the one-line reason each dies. Why is “concrete” the load-bearing qualifier in the no-impossibility claim? What single property does each prior-art construction give up?

5 Your first move

The work is **purely non-computational mathematics** from here — every .py is finitary because the finitary part is the whole computable boundary; the residue has no LP to run. Two branches, sharing one wall:

(a) The deep unlock

the σ -OML duality (descent axis). Build the Loomis–Sikorski-type representation / countable-join-preserving σ -Stone duality for a concrete non-Boolean infinite σ -OML, or prove none exists. High ceiling, **no current foothold**.

(b) The tractable special case

the singular sliver. Does a *singular* orthoadditive state on $L(H)$ extend to a charge on $S_0(L(H))$? A concrete operator-algebra question (Calkin, Bunce–Wright, Takesaki) about one lattice. **Conditionally** self-contained.

Do (b) first — it is informative either way. If a singular state fails to extend, then *no* state on $L(H)$ extends, closing the PR_{dual} “does-descent-do-independent-work” question negatively. If one *does* extend, it is the first concrete object that extends but may not concentrate — handing branch (a) its motivating example. (*Note §5.*)

The hinge that gates (b) — settle this first of all

Is the singular-extension obstruction reached with a **finite** or a **countable** meet-zero family? This is **OPEN** — and it decides whether (b) is genuinely the easier branch.

- The normal-state kill is finitary (lines in a 2-plane) and so dodges the σ -wall. For singular states the relevant families are *infinite-dimensional* subspaces with trivial intersection, so that *line-clustering* route is **vacuous** ($s(e) = 0$ on all finite-dim e).
- **But “the line route is vacuous” is NOT “the obstruction is countable.”** A *finite* family of infinite-dimensional pairwise-meet-zero subspaces with $\sum s(a_i) > 1$ is **unanalysed**. If such a finite obstruction exists, (b) is a self-contained operator-algebra problem you can settle ahead of (a). If only a countable family obstructs, (b) is entangled with the same finitary-to- σ bridge as (a) and the tractability advantage evaporates.
- (*Honesty note: a 2026-06-04 claim that this was “settled = infinitary, both branches gate on the σ -wall” was an over-read of lh_singular_dichotomy.md — that file calls the extension question OPEN. No new mathematics settled it. It is still open. This is exactly the kind of blocked-routes- $\neq$ -impossible slip the thread guards against for residue #2; don’t repeat it.*)

Concretely, the very first thing to try: take a *finite* family of infinite-dimensional, pairwise-trivially-intersecting subspaces of H and ask whether orthoadditivity of a singular state forces $\sum s > 1$ (finite obstruction) or whether singular states evade every finite bound (pushing the obstruction to countable). That single computation/argument decides the branch. (*Setup: verification/lh_singular_dichotomy.md lines 95–108 give the singular-state constructions and the open statement.*)

Checkpoint 5

State the finite-vs-countable hinge and what each outcome implies for branch (b). Why is (b) informative whether or not the singular state extends? What is the first concrete object to write down?

Methodological reminders (the thread’s hard-won rules)

- **Verify by building + reading the source**, not by grep or recollection. The session that produced this thread caught its central error (the (A)/(B) equivocation) and corrected a literature guess (HLS is σ -additive, not finite) only by reading primary sources in full.
- **Weaken “settled/strict/resolves” to “open/separation/conditional”** unless *both* directions are checked. The infinitary over-read above is the canonical cautionary case.
- **Blocked routes $\neq$ impossible.** Keep saying it.
- **Audit before drafting.** No LaTeX for unaudited claims.